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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- IF H is the harmonic mean between P and Q, then $\frac{H}{P} + \frac{H}{Q}$ is
(a) 2 (b) $\frac{P+Q}{Q}$ (c) $\frac{PQ}{P+Q}$ (d) None
- The number of values of k for which the system of equations $(k+1)x + 8y = 4k$ and $kx + (k+3)y = 3k-1$ has infinitely many solution, is
(a) 0 (b) 1 (c) 2 (d) infinite
- The sum of ${}^{20}C_8 + {}^{20}C_9 + {}^{21}C_{10} + {}^{22}C_{11} + {}^{23}C_{11}$ is
(a) ${}^{22}C_{12}$ (b) ${}^{23}C_{12}$ (c) 0 (d) ${}^{21}C_{10}$
- The value of $\cot^{-1}(21) + \cot^{-1}(13) + \cot^{-1}(-8)$ is
(a) 0 (b) π (c) ∞ (d) $\pi/2$
- Normal to the curve $y = x^3 - 3x + 2$ at that point (2, 4) is
(a) $9x - y - 14 = 0$ (b) $x - 9y + 40 = 0$
(c) $x + 9y - 38 = 0$ (d) $-9y + y + 22 = 0$
- The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \dots + \sin \frac{(n-1)\pi}{n} \right]$ is
(a) 0 (b) π (c) 2 (d) $\frac{\pi}{2}$
- The point on the curve $y = 6x - x^2$, where the tangent is parallel to X-axis is
(a) (0,0) (b) (2,8) (c) (6,0) (d) (3,9)
- If $I_1 = \int_0^1 2x^2 dx$, $I_2 = \int_0^1 2x^3 dx$, $I_3 = \int_1^2 2x^2 dx$ and $I_4 = \int_1^2 2x^3 dx$, then
(a) $I_1 = I_2$ (b) $I_2 > I_1$ (c) $I_3 > I_4$ (d) $I_4 > I_3$
- The value of integral $\int_0^{\pi/2} \log \tan x dx$ is
(a) π (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) 0
- A determinant is chosen at random from the set of all determinants of matrices of order 2 with elements 0 and 1 only. The probability that the determinant chosen is non-zero, is
(a) $\frac{3}{16}$ (b) $\frac{3}{8}$ (c) $\frac{1}{4}$ (d) None
- If $\sin^2 x = 1 - \sin x$, then $\cos^4 + \cos^2 x$ is equal to
(a) 0 (b) 1 (c) $\frac{2}{3}$ (d) -1
- The equation of the plane passing through the point (1, 2, 3) and having the vector $N = 3i + j + 2k$ as its normal, is
(a) $2x - y + 3z + 7 = 0$ (b) $3x - y + 2z + 7 = 0$
(c) $3x - y + 2z = 7$ (d) $3x + 2z = 7$
- The value of $\int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt$ is
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) 1 (d) None
- Coefficient of quadratic equation $ax^2 + bx + c = 0$ are chosen by tossing three fair coins, where 'head' means one and 'tail' means two. Then the probability that roots of the equation are imaginary, is
(a) $\frac{7}{8}$ (b) $\frac{5}{8}$ (c) $\frac{3}{8}$ (d) $\frac{1}{8}$
- In a class of 100 students, 55 students have passed in Mathematics and 67 students have passed in Physics. Then, the number of students who have passed in Physics only, is
(a) 22 (b) 33 (c) 10 (d) 45
- IF (4, -3) and (-9, 7) are the two vertices of a triangle and (1, 4) is its centroid, then area of triangle is
(a) $\frac{138}{2}$ (b) $\frac{319}{2}$ (c) $\frac{183}{2}$ (d) $\frac{381}{2}$
- The equation of the ellipse with major axis along the X-axis and passing through the points (4, 3) and (-1, 4) is
(a) $15x^2 + 7y^2 = 247$ (b) $7x^2 + 15y^2 = 247$
(c) $16x^2 + 9y^2 = 247$ (d) $9x^2 + 16y^2 = 247$
- If the circles $x^2 + y^2 + 2x + 2ky + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally, then k is
(a) 2 or $-\frac{3}{2}$ (b) -2 or $-\frac{3}{2}$ (c) 2 or $\frac{3}{2}$ (d) -2 or $\frac{3}{2}$
- Focus of the parabola $x^2 + y^2 - 2xy - 4(x + y - 1) = 0$ is
(a) (1, 1) (b) (1, 2) (c) (2, 1) (d) (0, 2)
- If a, b and c are unit vectors such that $a + b + c = 0$, then the value of $a \cdot b + b \cdot c + c \cdot a$ is
(a) $\frac{2}{3}$ (b) $-\frac{2}{3}$ (c) $\frac{3}{2}$ (d) $-\frac{3}{2}$
- IF two towers of heights h_1 and h_2 subtend angles 60° and 30° respectively at the mid-point of the line joining their feet, then $h_1 : h_2$ is
(a) 1:2 (b) 1:3 (c) 2:1 (d) 3:1
- If the vectors $a = (1, x, -2)$ and $b = (x, 3, -4)$ are mutually perpendicular, then value of x is
(a) -2 (b) 2 (c) 4 (d) -4
- What is the value of α for which $f(x) = \begin{cases} \sin x, & \text{if } x \leq \frac{\pi}{2} \\ ax, & \text{if } x > \frac{\pi}{2} \end{cases}$ is continuous?
(a) π (b) $\frac{\pi}{4}$ (c) $\frac{2}{\pi}$ (d) 0
- If the real number x when added to its inverse gives the minimum value of the sum, then the value of x is equal to
(a) -2 (b) 2 (c) 1 (d) -1
- If $\cos(\alpha + \beta) = \frac{4}{5}$ and $\sin(\alpha - \beta) = \frac{5}{13}$, $0 < \alpha, \beta, \frac{\pi}{4}$, then $\tan(2\alpha)$ is equal to
(a) $\frac{56}{33}$ (b) $\frac{63}{65}$ (c) $\frac{16}{63}$ (d) $\frac{33}{56}$
- The number of words that can be formed by using the letters of the word 'MATHEMATICS' that start as well as end with T is
(a) 80720 (b) 90720 (c) 20860 (d) 37528
- If $A - B = \frac{\pi}{4}$, then $(1 + \tan A)(1 - \tan B)$ is equal to
(a) 2 (b) 1 (c) 0 (d) 3
- Let P(E) denote the probability of event E. Given $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$, the values of $P(A|B)$ and $P(B|A)$ respectively are
(a) $\frac{1}{4}, \frac{1}{2}$
(b) $\frac{1}{2}, \frac{1}{4}$
(c) $\frac{1}{2}, 1$
(d) $1, \frac{1}{2}$
- The number of different license plates that can be formed in the format 3 English letters (A ... Z) followed by 4 digits (0, 1, ... 9) with repetitions allowed in letters and digits is equal to
(a) $26^3 \times 10^4$ (b) $26^3 + 10^4$
(c) 36 (d) 26^3
- Which of the following is correct?
(a) $\sin 1^\circ > \sin 1$ (b) $\sin 1^\circ = \sin 1 \sin 1^\circ = \frac{\pi}{180} \sin 1$
(c) $\sin 1^\circ < \sin 1$ (d) $\sin 1^\circ = \sin 1 \sin 1^\circ = \frac{\pi}{180} \sin 1$
- If a, b, c are non-coplanar vectors and λ is a real number, then the vectors $a + 2b + 3c$, $\lambda b + 4c$ and $(2\lambda - 1)c$ are non-coplanar for
(a) All values of λ
(b) All except one values of λ
(c) All except two values of λ
(d) No value of λ



32. Suppose values taken by a random variable X are such that $a \leq x_i \leq b$, where x_i denotes the value of X in the i th case for $i = 1, 2, 3, \dots, n$, then
- $(b - a)^2 \geq \text{Var}(X)$
 - $\frac{a^2}{4} \leq \text{Var}(X)$
 - $a^2 \leq \text{Var}(X) \leq b^2$
 - $a \leq \text{Var}(X) \leq b$
33. If ω is the cube root of unity, then the system of equations $x + \omega^2 y + \omega z = 0$, $\omega x + y + \omega^2 z = 0$ and $\omega^2 x + \omega y + z = 0$ is
- consistent and has unique solution
 - consistent and has more than one solution
 - inconsistent
 - None of the above
34. If $x = \log_a bc$, $y = \log_b ca$ and $z = \log_c ab$, then $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$ is equal to
- abc
 - $\sqrt{ab} + \sqrt{bc} + \sqrt{ca}$
 - 1
 - $x + y + z$
35. If $2^a = 3^b = 6^{-c}$, then $ab + bc + ca$ is equal to
- 1
 - 2
 - 0
 - None of these
36. If e and e' be the eccentricities of a hyperbola and its conjugate, then $\frac{1}{e^2} + \frac{1}{e'^2}$ is equal to
- 0
 - 1
 - 2
 - None of these
37. If a fair coin is tossed n times, then the probability that the head comes odd number of times is
- $\frac{1}{2}$
 - $\frac{1}{2^n}$
 - $\frac{1}{2^{n-1}}$
 - None of these
38. If $\sin(\pi \cos \theta) = \cos(\pi \sin \theta)$, then $\sin 2\theta$ is equal to
- $\pm \frac{3}{4}$
 - $\pm \frac{1}{3}$
 - $\pm \frac{1}{4}$
 - $\pm \frac{4}{3}$
39. In which of the following regular polygons, the number of diagonals is equal to number of sides?
- Pentagon
 - Square
 - Octagon
 - Hexagon
40. One hundred identical coins each with probability P of showing up heads are tossed. If $0 < P < 1$ and the probability of heads showing on 50 coins is equal to that of heads on 51 coins, then the value of P is
- $\frac{1}{2}$
 - $\frac{49}{101}$
 - $\frac{50}{101}$
 - $\frac{51}{101}$
41. The equation $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$, where x is a variable has real roots. Then, the interval of p is
- $(0, 2\pi)$
 - $(-\pi, 0)$
 - $(\frac{\pi}{2}, \frac{\pi}{2})$
 - $(0, \pi)$
42. Number of real roots of $3x^5 + 15x - 8 = 0$ is
- 3
 - 5
 - 1
 - 0
43. The value of k for which the set of equations $3x + ky - 2z = 0$, $x + ky + 3z = 0$ and $2x + 3y - 4z = 0$ has a non-trivial solution, is
- $\frac{15}{2}$
 - $\frac{17}{2}$
 - $\frac{31}{2}$
 - $\frac{33}{2}$
44. If $x = \log_3 5$, $y = \log_{17} 25$, then which one of the following is correct?
- $x > y$
 - $x < y$
 - $x \leq y$
 - $x = y$
45. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then A^n for any natural number n is
- $\begin{bmatrix} n & n \\ 0 & n \end{bmatrix}$
 - $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$
 - $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 - None of these
46. A problem in Mathematics is given to three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$, and $\frac{1}{4}$, respectively. If they all try to solve the problem, what is the probability that the problem will be solved?
- $\frac{1}{2}$
 - $\frac{1}{4}$
 - $\frac{1}{3}$
 - $\frac{3}{4}$
47. The function x^x decreases in the interval
- $(0, e)$
 - $(0, 1)$
 - $(0, \frac{1}{e})$
 - None of these



48. If $a + b + c = 0, |a| = 3, |b| = -5, |c| = 7$, then angles between the vectors a and b is
- $\frac{\pi}{2}$
 - $\frac{\pi}{3}$
 - $\frac{\pi}{4}$
 - $\frac{\pi}{6}$
49. If $\theta (0 \leq \theta \leq \pi)$ is the angle between the vectors a and b , then $\frac{|a \times b|}{a \cdot b}$ equal to
- $-\cot \theta$
 - $\tan \theta$
 - $-\tan \theta$
 - $\cot \theta$
50. If $f(a + b) = f(a) \times f(b)$ for all a and b and $f(5) = 2, f'(0) = 3$, then $f'(5)$ is equal to
- 2
 - 4
 - 6
 - 8

REASONING

51. If a man walks at the rate of 4 km/h, he misses a train by only 6 min. However, if he walks at the rate of 5 km/h he reaches the station 6 min before the arrival of the train. The distance covered by him to reach the station is
- 4 km
 - 7 km
 - 9 km
 - 5 km
52. The missing number in the given series 3, 6, 6, 12, 9, ..., 12 is
- 15
 - 18
 - 11
 - 13
53. A man runs 20 m towards east and turns right, runs 10 m and turns right, runs 9 m and turns left, runs 5 m and turns left, runs 12 m and finally turns left and runs, 6 m. Which direction is the man facing?
- North
 - South
 - East
 - West
54. In a club, there are certain number of males and females. If 15 females are absent, then number of males will be half of females. If 45 males are absent, then female strength will be 5 times that of males. Number of males actually present is
- 45
 - 80
 - 105
 - 175
55. The missing number in the following series 6, 12, 21, ..., 48 is
- 40
 - 33
 - 38
 - 45

Directions (Q.Nos. 56-58) Read the following passage carefully and answer the questions.

Six boys A, B, C, D, E and F are marching in a line. They are arranged according to their heights, the tallest being at the back and the shortest in the front. F is between B and A. E is shorter than D but taller than C who is taller than A. E and F have two boys between them. A is not shortest among them.

56. Where is E?
- Between A and B
 - Between C and A
 - Between D and C
 - In front of C
57. If we start counting from the shortest, which boy is fourth in the line?
- E
 - A
 - D
 - C
58. Who is next to the shortest?
- C
 - B
 - E
 - F
59. Let x, y and z be distinct integers, x and y are odd and positive and z is even and positive. Which one of the following statements cannot be true?
- $(x-z)^2 y$ is even
 - $(x-z) y^2$ is odd
 - $(x-z) y$ is odd
 - $(x-y)^2 z$ is even
60. Pointing to a man in the photograph a lady said, "The father of his brother is the only son of my mother." How is this man in photograph related to the lady?
- Brother
 - Son
 - Grandson
 - Nephew
61. Find the odd number in the following series. 2, 9, 28, 65, 126, 216, 344, ...
- 28
 - 65
 - 126
 - 216
62. Average age of students of an adult school is 40 yr. 120 new students whose average age is 32 yr joined the school. As a result the average age is decreased by 4 yr. The number of students of the school after joining of the new student is
- 1200
 - 120
 - 360
 - 240
63. The letters P, Q, R, S, T, U and V not necessarily in that order represent seven consecutive integers from 22 to 33 and
- U is as much less than Q as R is greater than S.
 - V is greater than U.
 - Q is the middle term.
 - P is greater than S.
- Then, the sequence of letters from the lowest value to the highest value, is
- TVPQRSU
 - TRSQUPV
 - TUSQRPV
 - TVPQSRU





64. The minimum number of tiles of size 16 by 24 required to form a square by placing them adjacent to one another is
(a) 6
(b) 8
(c) 11
(d) 16
65. Five persons K, L, M, N and O are sitting around a dining table. K is the mother of M, M is actually the wife of O, N is the brother of K and L is the husband of K. How is N related to L?
(a) Son
(b) Cousin
(c) Brother
(d) Brother-in-law
66. Three men A, B and C plays cards. If one loses the game he has to give ₹ 3. If he wins the game he will gain ₹ 3 each from the other two losers. If A has won 3 games, B loses ₹ 3, C wins ₹ 12, then the total number of games played is
(a) 12
(b) 21
(c) 20
(d) 6

Directions (Q. Nos. 67-69) Read the following passage carefully and answer the questions.

- A causes B or C but not both.
 - F occurs only if B occurs.
 - D occurs, if B or C occurs.
 - E occurs only if C occurs.
 - J occurs only if E or F occurs.
 - D causes G or H or both.
 - H occurs, if E occurs.
 - G occurs, if F occurs.
67. If A occurs, which may occurs?
(i) F and G
(ii) E and H
(iii) D
(a) Only I
(b) Only II
(c) I and III or II and III, but not both
(d) I, II and III
68. If B occurs, which must occur?
(a) D
(b) G
(c) H
(d) J
69. If J occurs, which must have occurred?
(a) Both E and F
(b) Either B or C
(c) Both B and C
(d) None of these
70. If 'ROAST' is coded as 'PQYUR' in a certain language, then 'SLOPPY' is coded in that language as
(a) MRNAQN
(b) NRMNQA
(c) QNMRNA
(d) RANNMO

71. If 'lelibroon' means 'yellow hat', 'plekafroti' means 'flower garden' and 'frotimix' means 'garden salad', then which word could mean 'yellow flower'?
(a) Lelifroti
(b) Lelipleka
(c) Plekabroon
(d) Frotibroon
72. If + is *, - is +, * is / and / is -, then $6-9+8*3/20$ is equal to
(a) -2
(b) 6
(c) 10
(d) 12
73. In a certain year, there were exactly four Fridays and four Mondays in January. On what day of the week did the 20th of January fall that year?
(a) Saturday
(b) Sunday
(c) Thursday
(d) Tuesday
74. Krishna said, "This girl is the wife of grandson of my mother". How is Krishna related to girl?
(a) Father
(b) Father-in-law
(c) Husband
(d) Grandfather
75. Instead of walking along two adjacent sides of a rectangular field, a boy took a shortcut along the diagonal of the field and saved a distance equal to half the longer side. The ratio of the shorter side of the rectangle to the longer side is
(a) $\frac{1}{2}$
(b) $\frac{2}{3}$
(c) $\frac{1}{4}$
(d) $\frac{3}{4}$
76. Each word in parenthesis below is formed in a method. This method is used in all four examples.
SNIP(NICE)PACE
TEAR(EAST)FAST
TRAY(RARE)FIRE
POUT(OURS)CARS
Based on this method, the word in the parenthesis of CANE(?)BATS is
(a) NEAT
(b) CATS
(c) ANTS
(d) NETS
77. A study of native born residents in an area of Adivasis found that two-third of the children developed considerable levels of the nearsightedness after starting school, while their illiterate parents and grandparents, who had no opportunity for formal schooling, showed no signs of this disability. If the above statements are true, which of the following conclusions is most strongly supported by them?
(a) Only people who have the opportunity for formal schooling develop nearsightedness
(b) People who are illiterate do not do not suffer from nearsightedness
(c) The nearsightedness in the children is caused by the visual stress required by reading and other class work
(d) Only literate people are nearsighted



Directions (Q.Nos.78-80) Read the following passage carefully and answer the questions.

Five roommates Randy, Sally, Terry, Uma and Vernon each do one housekeeping task mopping, sweeping, laundry, vacuuming or dusting one day a week, Monday through Friday.

- Vernon does not vacuum and does not do his task on Tuesday.
- Sally does the dusting and does not do it on Monday or Friday.
- The mopping is done on Thursday.
- Terry does his task, which is not vacuuming, on Wednesday.
- The laundry is done on Friday and not by Uma.
- Randy does his task on Monday.

78. The task done by Terry on Wednesday is

- Vacuuming
- Dusting
- Mopping
- Sweeping

79. The day on which the vacuuming is done, is

- Friday
- Monday
- Tuesday
- Wednesday

80. Sally does dusting on

- Friday
- Monday
- Tuesday
- Wednesday

Direction(Q.Nos. 81-82) Read the following passage carefully and answer the questions.

P,Q, R, S, T, U,V and W are sitting round the circle and are facing the centre. P is second to the right of T, T is the neighbour of R and V. S is not the neighbour of P,V is the neighbour of U,Q is not between S and W and W is not between U and S.

81. Which two of the following are not neighbours?

- RV
- UV
- RP
- QW

82. What is the position of S?

- Between U and V
- Second to the right of P
- To the immediate right of W
- Data inadequate

83. The ratio between a two-digit number and the sum of the digits of that number is 4 : 1. If the digit in the unit's place is 3 more than the digit in ten's place, then the number is

- 24
- 63
- 36
- 42

84. Two positions of a dice are shown below. When number 1 is in the top, What number will be at the bottom?



(i)



(ii)

- 2
- 3
- 5
- Cannot be determined

85. A, B, C, D, E, F and G are sitting in a line facing East. C is immediate to the right of D. B is at one of the extreme ends has E as his neighbour. G is between E and F.D is sitting third from the South end. Who is sitting third from North?

- A
- E
- F
- G

86. There is a family party consisting of two fathers, two mothers, two sons, one father-in-law, one mother-in-law, one daughter-in-law, one grandfather, one grandmother and one grandson. What is the minimum number of persons required, so that this is possible?

- 5
- 6
- 7
- 8

87. If A is brother of B, C is brother of B and A is brother of D, then which of the following must be true?

- A is brother of C
- B is brother of C
- D is brother of C
- B is brother of D

Directions(Q. Nos. 88-90) Read the following passage carefully and answer the questions.

Five houses lettered A, B, C, D and E are built in a row next to each other. The houses are lined up in the order A, B, C, D and E. Each of the five houses have coloured roofs and chimneys. The roof and chimney of each houses must be painted as follows.

- The roof must be painted either green, red or yellow.
- The chimney must be painted either white, black or red.
- No house may have the same colour chimney as the colour of roof.
- No house may use any of the same colours that adjacent house uses.
- House E has a green roof.
- House B has a red roof and a black chimney.

88. Which of the following is true?

- Atleast two houses have a black chimney.
- Atleast two houses have red roofs.
- Atleast two houses have white chimneys
- Atleast two houses have green roofs

89. If house C has a yellow roof, then which of the following must be true?

- House E has a white chimney
- House E has a black chimney
- House E has a red chimney
- House D has a red chimney

90. What is the maximum number of green roofs?

- 1
- 2
- 3
- 4

ENGLISH

91. For a word, four spellings are given. Choose the correct one.

- Ceiling
- Cealing
- Ceiling
- Ceeling





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92. Choose the wrongly spelt word.

- (a) Believe
- (b) Relieve
- (c) Grieve
- (d) Decieve

93. Choose the word or phrase that is most similar in meaning to the word POLEMIC.

- (a) Black
- (b) Magnetic
- (c) Grimace
- (d) Controversial

94. The sentence below has 2 blanks. Fill in the blanks picking the appropriate pair of words from the ones given below that best completes the meaning of the sentences.

The most technologically advanced societies have been responsible for the greatest....., indeed, savagery seems to be in direct proportion to.....

- (a) wars; viciousness
- (b) catastrophes; ill-will
- (c) atrocities; development
- (d) triumphs; civilisation

95. Fill in the blank with the correct form of tense. The thief ... before the police came.

- (a) escaped
- (b) had escaped
- (c) will escape
- (d) has been escaped

96. Fill in the blank with the with appropriate words given. Anne had to pay for everything because as usual, Peter... his wallet at home.

- (a) had left
- (b) was leaving
- (c) left
- (d) leave

97. Pick the synonym of the word 'Meagre'.

- (a) Helpful
- (b) Abundant
- (c) Essential
- (d) Limited

98. Choose the words that best express the meaning of the given idiom-Mud slinging.

- (a) Giving pain
- (b) Abusing someone
- (c) Laying blame
- (d) Damaging the reputation

99. Pick the antonym of the word 'Timid'

- (a) Bold
- (b) Lazy
- (c) Calm
- (d) Slow

100. Pick the part of the sentence that has an error. If you would have come to me, I would have helped you.

- (a) If you would have
- (b) Come to me
- (c) I would have
- (d) Helped you

101. Choose the word or phrase that is most nearly opposite in meaning to the word 'Extrinsic'.

- (a) Reputable
- (b) Inherent
- (c) Ambitious
- (d) Cursory

102. Select the alternative giving the closest meaning of the idiom- To eat a humble pie.

- (a) To become a vegetarian
- (b) Disinfecting everything
- (c) To fill one's belly
- (d) To say you are sorry for a mistake that you made

103. Pick the antonym of the word 'Fabricate'

- (a) Construct
- (b) Weaken
- (c) Dismantle
- (d) Evolve

Directions(Q nos. 104-110) Fill in the blank with correct option to make a proper sentence.

104. The people... you socialize are called friends.

- (a) with whom
- (b) who
- (c) with who
- (d) whom

105. to school yesterday?

- (a) Did you walk
- (b) Did you walked
- (c) Do you walk
- (d) Have you walked

106. There was no... in the railway compartment for additional passengers.

- (a) space
- (b) place
- (c) seat
- (d) room

107. And now for this evening's main headline; Britain...another olympic gold medal.

- (a) had won
- (b) wins
- (c) won
- (d) has won

108. If she about his financial situation, she would have helped him out.

- (a) knew
- (b) had been knowing
- (c) had known
- (d) have known

109. I am sure she can teach computers as well. She's not new to the subject.

- (a) all together
- (b) altogether
- (c) together
- (d) together

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110. You are trying to drag me...a controversy.

- (a) in
- (b) into
- (c) from
- (d) for

COMPUTER

111. An I/O processor controls the flow of information between

- (a) cache memory and I/O devices
- (b) main memory and I/O devices
- (c) two I/O devices
- (d) cache and main memories

112. Which of the following devices will take highest time in taking the backup of the data from a computer?

- (a) Magnetic disk
- (b) Pen drive
- (c) CD
- (d) Magnetic tape

113. ROM is a kind of

- (a) primary memory
- (b) cache memory
- (c) removable memory
- (d) secondary memory

114. The errors that can be pointed out by compilers are

- (a) Syntax errors
- (b) Semantic errors
- (c) Logical errors
- (d) Internal errors

115. Let $x = 11111010$ and $y = 00001010$ be two 8-bit 2's complement numbers. Their product in 2's complement notation is

- (a) 11000100
- (b) 10011100
- (c) 10100101
- (d) 11010101

116. The range of numbers that can be stored in 8 bits, if negative numbers are stored in 2's complement form is

- (a) -128 to + 128
- (b) -127 to + 128
- (c) -128 to + 127
- (d) -127 to + 127

117. Primary storage is ... as compared to secondary memory.

- (a) slow and expensive
- (b) fast and inexpensive
- (c) fast and expensive
- (d) slow and inexpensive

118. Which of the following units is used to supervise each instruction in the CPU?

- (a) Control unit
- (b) Accumulator
- (c) ALU
- (d) Control Register

119. $(2FAOC)_{16}$ is equivalent to

- (a) $(195084)_{10}$
- (b) $(00101111101000001100)_2$
- (c) Both (a) and (b)
- (d) None of the above

120. The octal equivalent of binary number 111010 is

- (a) 81
- (b) 72
- (c) 71





MAARULA CLASSES

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ANSWER KEY

1	A	2	B	3	C	4	B	5	C
6	A	7	D	8	D	9	D	10	B
11	B	12	C	13	A	14	A	15	D
16	C	17	B	18	A	19	A	20	D
21	B	22	A	23	C	24	B	25	A
26	B	27	A	28	D	29	A	30	B
31	C	32	A	33	B	34	C	35	C
36	B	37	A	38	A	39	A	40	D
41	D	42	C	43	D	44	A	45	B
46	D	47	C	48	B	49	C	50	C
51	A	52	B	53	A	54	B	55	B
56	C	57	D	58	D	59	A	60	D
61	D	62	D	63	C	64	A	65	D
66	A	67	C	68	B	69	B	70	C
71	B	72	C	73	B	74	B	75	D
76	C	77	C	78	D	79	B	80	C
81	A	82	D	83	C	84	C	85	D
86	A	87	A	88	B	89	A	90	C
91	C	92	D	93	D	94	C	95	B
96	C	97	D	98	D	99	A	100	A
101	B	102	D	103	A	104	A	105	A
106	D	107	B	108	D	109	B	110	A
111	B	112	D	113	D	114	A	115	A
116	B	117	C	118	A	119	C	120	b



MATHS SOLUTION

1. (a) H is the harmonic mean between P and Q (given)

$$H = \frac{2PQ}{P+Q} \Rightarrow \frac{H}{2} = \frac{PQ}{P+Q}$$

$$\frac{H}{P} + \frac{H}{Q} = H \left(\frac{P+Q}{PQ} \right) = H \left(\frac{2}{H} \right) = 2$$

2. (b) Condition that 2 lines are coincident

$$a_1x + b_1y + c_1 = 0 \Rightarrow a_2x + b_2y + c_2 = 0$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow (k+1)x + 8y = 4k$$

$$\Rightarrow kx + (k+3)y = 3k - 1 \Rightarrow \frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1}$$

$$k^2 + 4k + 3 = 8k; 24k - 8 = 4k^2 + 12k$$

$$k^2 - 4k + 3 = 0; 4k^2 - 12k + 8 = 0$$

$$(k-3)(k-1) = 0; k^2 - 3k + 2 = 0$$

$$k = 3, k = 1 \text{ and } k = 1, 2.$$

$\therefore$ For $k = 1$ the lines will have infinite solution.

$\therefore$ Number of values of $k = 1$.

3. (c) $({}^{20}C_9 + {}^{20}C_8) + {}^{21}C_{10} + {}^{22}C_{11} - {}^{23}C_{11}$

$$= ({}^{21}C_9 + {}^{21}C_{10}) + {}^{22}C_{11} - {}^{23}C_{11}$$

$$[\because {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r]$$

$$({}^{22}C_{10} + {}^{22}C_{11}) - {}^{23}C_{11} = {}^{23}C_{11} - {}^{23}C_{11} = 0$$

4. (b)

$$\therefore \left[\sin^{-1} \left(\frac{2x}{1+x^2} \right) = 2\tan^{-1} x, \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = 2\tan^{-1} x \right]$$

$$\tan^{-1} \left(\frac{2x}{1-x^2} \right) = 2\tan^{-1} x$$

$$2\tan^{-1} a - 2\tan^{-1} b = 2\tan^{-1} x$$

$$\Rightarrow \tan^{-1} \left(\frac{a-b}{1+ab} \right) = \tan^{-1} x \Rightarrow x = \frac{a-b}{1+ab}$$

$$\cot^{-1}(21) + \cot^{-1}(13) + \cot^{-1}(-8) = ?$$

$$= \tan^{-1} \left(\frac{1}{21} \right) + \tan^{-1} \left(\frac{1}{13} \right) + \tan^{-1} \left(-\frac{1}{8} \right)$$

$$= \tan^{-1} \left(\frac{\frac{1}{21} + \frac{1}{13}}{1 - \frac{1}{21 \cdot 13}} \right) + \cot^{-1}(-8)$$

$$= \tan^{-1} \frac{1}{8} + \pi - \cot^{-1} 8 [\because \cot^{-1}(-x) = \pi - \cot^{-1} x]$$

$$= \cot^{-1} 8 + \pi - \cot^{-1} 8 = \pi$$

5. (c) Normal to the curve $y = x^3 - 3x + 2$ at point $(2, 4)$

$$y = x^3 - 3x + 2$$

$$\frac{dy}{dx} = 3x^2 - 3$$

$$\left. \frac{dy}{dx} \right|_{(2,4)} = 3 \times 2^2 - 3 = 9$$

$$\text{Slope of tangent} = 9 = m_1$$

$$\text{Now, slope of normal } (m_2) = -1/\text{Slope of tangent } (m_1)$$

$$m_2 = \frac{-1}{9}$$

$$\text{Equation of normal at } (2, 4)$$

$$y - 4 = \frac{-1}{9}(x - 2)$$

$$\Rightarrow 9y - 36 = -x + 2$$

$$\Rightarrow x + 9y = 38$$

$$\Rightarrow x + 9y - 38 = 0$$

6. (a)

$$\lim_{n \rightarrow \infty} \frac{\pi}{n} \left[\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \dots + \sin \frac{(n-1)\pi}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{\pi}{n} \left\{ \sin \left(\frac{\pi}{n} \right) + \sin \left(\frac{\pi}{n} + \frac{\pi}{n} \right) + \sin \left(\frac{\pi}{n} + \frac{2\pi}{n} \right) \right\} + \dots + \sin \left(\frac{\pi}{n} + \frac{n\pi}{n} \right)$$

$$\therefore \sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \dots + \sin(\alpha + n\beta) = \frac{\sin \left(\frac{2\alpha + n\beta}{2} \right) \sin \frac{n\beta}{2}}{\sin \frac{\beta}{2}}$$

$$= \lim_{n \rightarrow \infty} \frac{\pi}{n} \frac{\sin \left(\frac{\pi}{n} \left(\frac{\pi}{n} + \frac{n\pi}{n} \right) \right) \sin \frac{n}{2} \cdot \frac{\pi}{n}}{\sin \frac{\pi}{2n}}$$

$$= \lim_{n \rightarrow \infty} \frac{\pi}{n} \frac{\sin \left(\frac{2n + n\pi}{n} \right) \sin \frac{\pi}{2n}}{\sin \frac{\pi}{2n}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2 \left(\frac{\sin \pi}{2n} \right)} \sin \left(\pi + \frac{2\pi}{n} \right) \cdot 1 \quad \left(\because \lim_{n \rightarrow \infty} \frac{\sin \frac{1}{\theta}}{\frac{1}{\theta}} = 1 \right)$$

$$= \frac{1}{2.1} \sin(\pi + 0)$$

$$= \frac{1}{2} \cdot 0 = 0$$

7. (d) The point on the curve $y = 6x - x^2$, where tangent is parallel to X -axis.

Any line parallel to X -axis, then slope of the line $m = 0$

$$y = 6x - x^2$$

$$\frac{dy}{dx} = 6 - 2x = 0$$

$$6 = 2x \Rightarrow x = 3$$

(because of parallel to X -axis)

Now, put $x = 3$ at the curve,

$$y = 6 \times 3 - 9 = 9 \text{ (point on the curve} = (3, 9))$$

8. (d) $I_1 = \int_0^1 2x^2 dx, I_2 = \int_0^1 2x^3 dx, I_3 = \int_1^2 2x^2 dx$

$$\text{and } I_4 = \int_1^2 2x^3 dx$$

$$0 < x < 1 \Rightarrow x^2 > x^3$$

$$\int_0^1 2x^2 > \int_0^1 2x^3 \Rightarrow I_1 > I_2$$

$$1 < x < 2 \Rightarrow x^3 > x^2$$

$$\int_1^2 2x^3 > \int_1^2 2x^2 \Rightarrow I_4 > I_3$$

9. (d) $I = \int_0^{\pi/2} \log \tan x dx$

$$I = \int_0^{\pi/2} \log \tan \left(\frac{\pi}{2} - x \right) dx$$

$$I = \int_0^{\pi/2} \log \cot x dx$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi/2} (\log \tan x + \log \cot x) dx$$

$$= \int_0^{\pi/2} \log \tan x \times \cot x dx = \int_0^{\pi/2} (\log 1) dx = 0$$

10. (b) A determinant of order 2 is of the form $\Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

Its value is $ad - bc$

The total number of ways of choosing a, b, c and d is

$$2 \times 2 \times 2 \times 2 = 16$$

If $\Delta \neq 0$

If either $ad = 1$ and $bc = 0$ or $ad = 0$ and $bc = 1$

But $ad = 1$ and $bc = 0$ if and only if $a = d = 1$ and at least one of b, c is zero.

Thus, $ad = 1$ and $bc = 0$ in three cases $[(a, b, c, d)]$

$$= (1, 0, 0, 1)(1, 0, 1, 1)(1, 1, 0, 1).$$

Similarly $ad = 0$ and $bc = 1$ in three cases $[(a, b, c, d)]$

$$= (0, 1, 1, 0)(1, 1, 1, 0), (0, 1, 1, 1).$$

Thus, the probability of the required event is $\frac{6}{16} = \frac{3}{8}$

11. (b)

$$\text{If } \sin^2 x = 1 - \sin x$$





$$\Rightarrow \sin x = 1 - \sin^2 x \Rightarrow \sin x = \cos^2 x$$

$$\text{Hence, } \cos^4 x + \cos^2 x = (\cos^2 x)^2 + \cos^2 x$$

$$= \sin^2 x + \cos^2 x = 1$$

- 12. (c)** We know equation of plane in 3D is-

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

Where $a, b, c \Rightarrow$ Direction ratios of vector normal to the plane

As $N = 3\hat{i} - \hat{j} + 2\hat{k}$ is normal to plane

Plane passing through the point $(1, 2, 3)$

$$\text{So, } 3(x - 1) + (-1)(y - 2) + 2(z - 3) = 0$$

$$\Rightarrow 3x - y + 2z - 3 + 2 - 6 = 0$$

$$\Rightarrow 3x - y + 2z - 7 = 0 \Rightarrow 3x - y + 2z = 7$$

- 13. (a)** $I = \int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt$

Check it for $x = \frac{\pi}{4}$

$$\int_0^{\frac{1}{2}} \sin^{-1} \sqrt{t} dt + \int_0^{\frac{1}{2}} \cos^{-1} \sqrt{t} dt$$

$$= \int_0^{\frac{1}{2}} [(\sin^{-1} \sqrt{t}) + (\cos^{-1} \sqrt{t})] dt = \int_0^{\frac{1}{2}} \frac{\pi}{2} dt = \frac{\pi}{4}$$

- 14. (a)** For imaginary roots $b^2 - 4ac < 0$

$$\text{Total cases} = 2 \times 2 \times 2 = 8$$

For $b = 1$ and a and c which can be chosen in 4 ways

For $b = 2$ either $a = 1, c = 2$ or $a = 2, c = 1$ or $a = 2, c = 2$

Sample space = $\{(1, 2, 2)(1, 1, 2)(1, 2, 1)(1, 1, 1)$

$(2, 1, 1)(2, 2, 1)(2, 1, 2)(2, 2, 2)\}$

Total number of favourable cases = 7

$$\text{Required probability} = \frac{7}{8}$$

- 15. (d)** Since, $n(M \cup P) = 100, n(M) = 55, n(P) = 67$

$$n(M \cup P) = n(M) + n(P) - n(M \cap P)$$

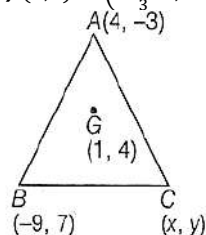
$$100 = 55 + 67 - n(M \cap P)$$

$$n(M \cap P) = 22$$

$$\text{Passed in Physics only} = n(P) - n(M \cap P)$$

$$= 67 - 22 = 45$$

- 16. (c)** $(1, 4) = \left(\frac{4-9+x}{3}, \frac{-3+7+y}{3}\right)$



$$x = 8, y = 8$$

$$\text{area}(\triangle ABC) = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$\text{area}(\triangle ABC) = \frac{1}{2} \begin{vmatrix} 4 & -3 & 1 \\ -9 & 7 & 1 \\ 8 & 8 & 1 \end{vmatrix} = \frac{183}{2}$$

- 17 (b)** Let equation of ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

($\because$ major axis is along X-axis)

Now, $(4, 3)$ and $(-1, 4)$ lies on the curve.

$$\therefore S_{(4,3)}: \frac{16}{a^2} + \frac{9}{b^2} = 1 \quad \dots\dots\dots (i)$$

$$S_{(-1,4)}: \frac{1}{a^2} + \frac{16}{b^2} = 1 \quad \dots\dots\dots (ii)$$

$$\text{Eq. (i)} - 16 \times \text{Eq. (ii)}$$

$$b^2 = \frac{247}{15} \text{ and } a^2 = \frac{247}{7}$$

$$\therefore \text{Required equation } 7x^2 + 15y^2 = 247$$

- 18. (a)**

$$x^2 + y^2 + 2x + 2ky + 6 = 0, x^2 + y^2 + 2ky + k = 0$$

$$C_1(-1, -k)$$

$$r_1 = \sqrt{1 + k^2 - 6} = \sqrt{k^2 - 5}$$

$$C_2(0, -k)$$

$$r_2 = \sqrt{k^2 - k}$$

For orthogonal intersection

$$r_1^2 + r_2^2 = (C_1 C_2)^2$$

$$(k^2 - 5) + (k^2 - k) = 1$$

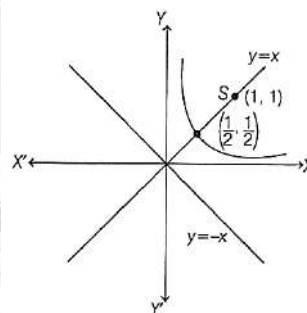
$$\Rightarrow 2k^2 - k - 6 = 0$$

$$\Rightarrow 2k^2 - 4k + 3k - 6 = 0$$

$$\Rightarrow 2k(k - 2) + 3(k - 2) = 0$$

$$\Rightarrow k = \frac{-3}{2}, k = 2$$

- 19. (a)** $x^2 + y^2 - 2xy - 4(x + y - 1) = 0$



$$\Rightarrow (x - y)^2 = 4\{(x + y) - 1\} \Rightarrow \frac{(x - y)^2}{2} = \frac{4(x + y - 1)}{2}$$

$$\left(\frac{x - y}{\sqrt{2}}\right)^2 = \frac{2\sqrt{2}(x + y - 1)}{\sqrt{2}}$$

Now, we can assume it as $Y^2 = 2\sqrt{2}X$

For finding the vertex $X = 0$ and $Y = 0$

Here,

$$x - y = 0 \quad \dots\dots\dots (i)$$

And

$$x + y = 1 \quad \dots\dots\dots (ii)$$

$$\text{On solving, we get } x = y = \frac{1}{2}$$

$$\therefore \text{Vertex of parabola} = \left(\frac{1}{2}, \frac{1}{2}\right)$$

The focus is the point of intersection of $X = a$ and $Y = 0$

$$a = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$$

$$\text{Hence, } \frac{x + y - 1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \text{ and } x - y = 0$$

Solving these two equations we will get focus $(1, 1)$.

- 20. (d)** $|a + b + c|^2 = |a|^2 + |b|^2 + |c|^2 + 2(a.b + b.c + c.a)$

$$0 = 1 + 1 + 1 + 2(a.b + b.c + c.a)$$

$$a.b + b.c + c.a = -\frac{3}{2}$$

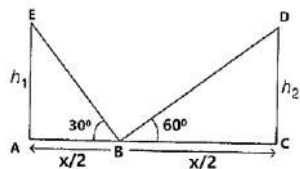
- 21 (b)**

In Triangle ABE

$$\tan 30^\circ = \frac{h_1}{x/2} = \frac{1}{\sqrt{3}}$$

$$x = 2\sqrt{3}h_1$$





And in a triangle BCD

$$\tan 60^\circ = \frac{h_2}{x/2} = \sqrt{3}$$

$$x = \frac{2h_2}{\sqrt{3}}$$

From Eqs. (i) and (ii), we get

$$2\sqrt{3}h_1 = \frac{2h_2}{\sqrt{3}}$$

$$\Rightarrow \frac{h_1}{h_2} = \frac{1}{3} = h_1:h_2 = 1:3$$

22. (a) For a and b to be perpendicular $a \cdot b = 0$

$$(\hat{i} + x\hat{j} - 2\hat{k}) \cdot (x\hat{i} + 3\hat{j} - 4\hat{k}) = 0$$

$$\Rightarrow x + 3x + 8 = 0 \Rightarrow 4x = -8 \Rightarrow x = -2$$

23. (c)

$$f(x) = \begin{cases} \sin x, & x \leq \frac{\pi}{2} \\ ax, & x > \frac{\pi}{2} \end{cases}$$

$$\text{LHL} = \lim_{x \rightarrow (\frac{\pi}{2})^-} \sin x = \sin \frac{\pi}{2} = 1$$

$$\text{RHL} = \lim_{x \rightarrow (\frac{\pi}{2})^+} ax = a \left(\frac{\pi}{2} \right)$$

Function is continuous at $x = \frac{\pi}{2}$.

Hence, LHL = RHL

$$\Rightarrow a \times \frac{\pi}{2} = 1 \Rightarrow a = \frac{2}{\pi}$$

24. (b)

by Given condition we get

$$\text{Let } f(x) = x + \frac{1}{x} \quad \dots\dots(i)$$

On differentiating w.r.t. x. we get

$$f'(x) = 1 - \frac{1}{x^2}$$

For max or min of $f(x)$

$$\text{Put } f'(x) = 0$$

$$\Rightarrow 1 - \frac{1}{x^2} = 0$$

$$\Rightarrow \frac{(x^2-1)}{x^2} = 0 \quad (\because x \neq 0)$$

$$\Rightarrow (x-1)(x+1) = 0$$

$$\Rightarrow x = 1 \text{ or } -1$$

$$\text{Now, } f''(x) = \frac{2}{x^3}$$

$$\text{At } x = -1 \quad f''(-1) = 2 \quad (\text{max})$$

$$\text{At } x = 1 \quad f''(1) = 2 \quad (\text{min})$$

So, $f(x)$ is min at $(x=1)$ and its minimum value at $(x=1)$ is

$$f(1) = 1 + \frac{1}{1} = 2$$

$$\text{Or Let } f(x) = x + \frac{1}{x}$$

$$\because \text{AM} \geq \text{GM}$$

$$\Rightarrow \frac{x+1}{2} \geq \left(x \cdot \frac{1}{x}\right)^{1/2} \Rightarrow \left(x + \frac{1}{x}\right) \geq 2$$

Min of $f(x)$ is 2.

25. (a)

$$\tan 2\alpha = \tan\{(\alpha + \beta) + (\alpha - \beta)\}$$

$$\tan 2\alpha = \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta)\tan(\alpha - \beta)}$$

$$\cos(\alpha + \beta) = \frac{4}{5}, \sin(\alpha - \beta) = \frac{5}{13}$$

$$\text{Hence, } \tan(\alpha + \beta) = \frac{3}{4}, \tan(\alpha - \beta) = \frac{5}{12}$$

$$\tan 2\alpha = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \cdot \frac{5}{12}} = \frac{56}{33}$$

$$\Rightarrow \tan 2\alpha = \frac{56}{33}$$

26. (b)

T										T
---	--	--	--	--	--	--	--	--	--	---

Total number of letters in the word "MATHEMATICS" is 11 and there are 2 T's, 2 A's and 2 M's

Since T is fixed at starting and at the ending,

$\therefore$ Required number of ways

$$= \frac{9!}{2!2!} = \frac{362880}{2 \cdot 2} = 90720$$

27. (a)

$$A - B = \frac{\pi}{4}$$

$$\tan(A - B) = \tan \frac{\pi}{4} \Rightarrow \frac{\tan A - \tan B}{1 + \tan A \tan B} = 1$$

$$\tan A - \tan B - \tan A \tan B = 1$$

$$\text{Now, } (1 + \tan A)(1 - \tan B)$$

$$= 1 - \tan B + \tan A - \tan A \tan B$$

$$= 1 + (\tan A - \tan B - \tan A \tan B) = 1 + 1 = 2$$

28. (d) $P(A) = 1, P(B) = \frac{1}{2}$ as $P(A) = 1$ it is certain event

$$\text{hence } P(A \cap B) = P(B)$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)}{P(A)} = \frac{1/2}{1} = \frac{1}{2}$$

29. (a) The number of arrangements of 3 English letters with repetitions allowed = $26 \cdot 26 \cdot 26 = (26)^3$

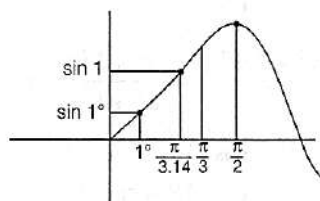
The number of arrangements of 4 digits with repetition allowed = $10 \cdot 10 \cdot 10 \cdot 10 = (10)^4$

$\therefore$ Required number of different licence plates

$$= (26)^3 \times (10)^4 \text{ (According to given option)}$$

30. (b)

$$\text{We should know, } 1 \cong \frac{\pi}{3.14}$$



From graph we can say $\sin 1 > \sin 1^\circ$

31. (c) For a and b to be perpendicular, $a \cdot b = 0$

$$\text{Let } a = \hat{i}, b = \hat{j}, c = \hat{k}$$

$$\text{For non-coplanar vectors, } \begin{vmatrix} 1 & 2 & 3 \\ 0 & \lambda & 4 \\ 0 & 0 & 2\lambda - 1 \end{vmatrix} \neq 0$$

This must be true

$$\lambda(2\lambda - 1) - 0 \neq 0 \Rightarrow \lambda(2\lambda - 1) \neq 0 \Rightarrow 0 \text{ and } \lambda = \frac{1}{2}$$

$$\lambda \in \mathbb{R} \text{ except true values i.e. } 0, 1/2.$$

32. (a) $a \leq x_i \leq b$

$$\text{Range} = \text{Maximum} - \text{Minimum} = b - a$$

We should know Standard deviation $\leq$ Range

$$\sigma \leq (b - a)$$

$$\text{Square both side } \sigma^2 \leq (b - a)^2$$

We should know the variance = (Standard deviation)²

33. (b) It is system of homogeneous equation.

$$\Delta = \begin{vmatrix} 1 & \omega^2 & \omega \\ \omega & 1 & \omega^2 \\ \omega^2 & \omega & 1 \end{vmatrix}$$

$$C_1 \rightarrow C_1 + C_2 + C_3$$





$$\Delta = \begin{vmatrix} 1+\omega+\omega^2 & \omega^2 & \omega \\ 1+\omega+\omega^2 & 1 & \omega^2 \\ 1+\omega+\omega^2 & \omega & 1 \end{vmatrix} = 0 \quad [\because 1+\omega+\omega^2=0]$$

Here $\Delta = 0$ it means system has non trivial solution (infinite solution)

So, the system of equation is consistent has more than one solution.

34.(c)
$$\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$$

$$= \frac{1}{1+\log_a bc} + \frac{1}{1+\log_b ca} + \frac{1}{1+\log_c ab}$$

$$= \frac{1}{\log_a a + \log_a bc} + \frac{1}{\log_b b + \log_b ca} + \frac{1}{\log_c c + \log_c ab}$$

$$= \frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc}$$

$$= \log_{abc} a + \log_{abc} b + \log_{abc} c = \log_{abc} abc = 1$$

35. (c) Let $2^a = 3^b = 6^{-c} = k$

$$\Rightarrow 2^a = k, 3^b = k, 6^{-c} = k$$

$$\Rightarrow 2 = k^{\frac{1}{a}}, 3 = k^{\frac{1}{b}}, 6 = k^{\frac{1}{c}} \Rightarrow 2 \times 3 = k^{\frac{1}{c}}$$

$$\Rightarrow k^{\frac{1}{a}} \times k^{\frac{1}{b}} = k^{\frac{1}{c}}$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} = \frac{1}{c} \Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0 \Rightarrow \frac{ab+bc+ca}{abc} = 0$$

$$\Rightarrow ab+bc+ca=0$$

36. (b) $\rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ Hyperbola

$e' \rightarrow \frac{-x^2}{a^2} + \frac{y^2}{b^2} = 1$ Conjugate hyperbola

$$e = \sqrt{1 + \frac{b^2}{a^2}}, e' = \sqrt{1 + \frac{a^2}{b^2}}$$

$$\frac{1}{e^2} + \frac{1}{e'^2} = \frac{1}{1+\frac{b^2}{a^2}} + \frac{1}{1+\frac{a^2}{b^2}} = \frac{a^2}{a^2+b^2} + \frac{b^2}{a^2+b^2}$$

$$= \frac{a^2+b^2}{a^2+b^2} = 1 \quad \frac{1}{e^2} + \frac{1}{e'^2} = 1$$

37. (b) There are two straight lines for infinite solutions these straight lines must be coincident

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1}$$

Solving these three equations we get $k = 1, 3$ But only $k = 1$ satisfy all the equations.

38. (a)

$$\sin(\pi \cos \theta) = \sin\left(\frac{\pi}{2} - \pi \sin \theta\right) \Rightarrow \pi \cos \theta = \frac{\pi}{2} - \pi \sin \theta$$

$$\Rightarrow \cos \theta = \frac{1}{2} - \sin \theta \Rightarrow \sin \theta + \cos \theta = \frac{1}{2}$$

$$\cos\left(\frac{\pi}{2} - \pi \cos \theta\right) = \cos(\pi \sin \theta)$$

$$\frac{\pi}{2} - \pi \cos \theta = \pm \pi \sin \theta$$

$$\frac{\pi}{2} - \pi \cos \theta = -\pi \sin \theta \Rightarrow \frac{1}{2} = \cos \theta - \sin \theta$$

$$1 - \sin 2\theta = \frac{1}{4} \Rightarrow \sin 2\theta = \frac{3}{4} \text{ and } \frac{\pi}{2} - \pi \cos \theta = \pi \sin \theta$$

$$\Rightarrow (\sin \theta + \cos \theta) = \frac{1}{2} \Rightarrow \sin \theta + \cos \theta = \frac{1}{2}$$

$$1 + \sin 2\theta = \frac{1}{4} \Rightarrow \sin 2\theta = -\frac{3}{4}$$

39. (a) Let number of sides in the polygon be n then number of its diagonal will be $\frac{n(n-3)}{2}$

$$\text{But it is given that, } n = \frac{n(n-3)}{2}$$

$$\Rightarrow 2n = n(n-3)$$

$$\Rightarrow 2 = n-3 \Rightarrow n = 5$$

Hence polygon will be pentagon.

40. (d)

By condition using binomial distribution

$${}^{100}C_{50}P^{50}(1-P)^{50} = {}^{100}C_{51}P^{51}(1-P)^{49}$$

$$\Rightarrow \frac{100!}{50!50!}(1-P) = \frac{100!}{51!49!}$$

$$\Rightarrow \frac{1}{50}(1-P) = \frac{P}{51}$$

$$\Rightarrow 51-51P = 50P$$

$$\Rightarrow 101P = 51$$

$$\text{So } P = \frac{51}{101}$$

41. (d) $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$

coefficient of $x^2 = a \neq 0$

$$\cos p - 1 \neq 0 \Rightarrow \cos p \neq 1 \Rightarrow p \neq 0, 2\pi$$

$$\text{Real roots ; } \Delta \geq 0 \Rightarrow \cos^2 p - 4(\cos p - 1)\sin p \geq 0$$

$$\cos^2 p - (4\sin p)\cos p + 4\sin p \geq 0$$

$$\text{Real roots ; } \Delta \geq 0 \Rightarrow \cos^2 p - 4(\cos p - 1)\sin p \geq 0$$

$\Rightarrow$ This is quadratic in $\cos p$.

$$\Rightarrow \cos^2 p - (4\sin p)\cos p + 4\sin p \geq 0 \text{ to be positive always if } D \leq 0$$

$$\Rightarrow (-4\sin p)^2 - 4(1)(4\sin p) \leq 0$$

$$\sin^2 p - \sin p \leq 0 \Rightarrow \sin p(\sin p - 1) \leq 0$$

$$\text{as we know } -1 \leq \sin p \leq 1 \text{ hence, } \sin p - 1 \leq 0 \forall p$$

Hence, $\sin p \geq 0$

$$\therefore \sin p \in [0, 1]$$

$$\Rightarrow p \in [0, \pi]$$

But $p \neq 0$ hence, $p \in (0, \pi]$

From given options $p \in (0, \pi)$

42. (c) $f(x) = 3x^5 + 15x - 8 = 0$

$$\Rightarrow f'(x) = 3(5x^4) + 15(1) + 0 \Rightarrow 15(x^4 + 1) > 0 \forall x \in \mathbb{R}$$

$$\therefore f'(x) > 0 \forall x \in \mathbb{R}$$

$\Rightarrow f(x)$ is an increasing function



We should know that range of odd degree polynomials are all real number.

Hence, $f(x)$ cuts at x -axis only at one point.

$\Rightarrow$ Number of real solution = 1

43 (d) It is system of homogeneous equation for non trivial solution (infinite solution) $\Delta = 0$.

$$\begin{vmatrix} 3 & k & -2 \\ 1 & k & 3 \\ 2 & 3 & -4 \end{vmatrix} = 0$$

$$= 3(-4k-9) - k(-4-6) - 2(3-2k) = 0$$

$$\Rightarrow k = \frac{33}{2}$$



- 9554548576**



61. (d)
- $$\begin{array}{ccccccc} 2 & 9 & 28 & 65 & 126 & 216 & 344 \dots\dots \\ \hline 2^3+1 & 3^3+1 & 4^3+1 & 5^3+1 & 6^3+1 & 7^3+1 & \end{array}$$
- or $2 = 1^3 + 1, \quad 9 = 2^3 + 1, \quad 28 = 3^3 + 1,$
 $65 = 4^3 + 1, \quad 126 = 5^3 + 1$
 and $344 = 7^3 + 1$
 But $216 = 6^3 + 0$ which is odd number among them.

62. (d) Let the total number of students before joining new students = x .
 $\Rightarrow$ After joining new 120 students = $x + 120$
 Now, by condition,
 $x \times 40 + 120 \times 32 = (x + 120) \times 36$
 $\Rightarrow 40x + 3840 = 36x + 4320$
 $\Rightarrow 4x = 480$
 $\therefore x = 120$
 $\therefore$ Total number of students = $x + 120 = 120 + 120 = 240$

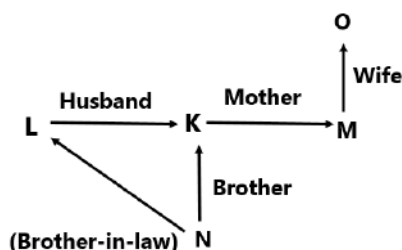
33. (c) By given condition, we get the required order (sequence) of letters from the lowest value to the highest value is
 $T < U < S < Q < R < P < V$
 i.e., TUSQRPV

64. (a) From option (a),
 Let the number of tiles = 6
 $\therefore$ Total length = 48
 and total breadth = 48



Since, length = breadth
 $\therefore$ Number of tiles form a square = 6

65. (d)



66. (a) Required total number of games played is 12.

Solutions (Q.Nos. 67-69)

- (i) A causes B or C but not both.
 (ii) F occurs only if B occurs.
 (iii) D occurs if B or C occurs.
 (iv) E occurs only if C occurs.
 (v) C occurs only if E or F occurs
 (vi) D causes G or H or both
 (vii) H occurs if E occurs.
 (viii) G occurs if F occurs.

67. (c) From Statement (i), A causes B or C but not both. From Statement (ii), F occurs only if B occurs and from Statement (iii), D occurs if B or C occur. It means I and II may occur. From Statements (vi) and (vii), II and III are may occur. So, we conclude that I and III or II and III may occur but not both occur.
68. (b) From Statement (ii) that F occurs only if B occurs and from Statement (viii) that if G occurs if F occurs it means if B occurs G must occur.
69. (b) From Statement (v), that J occurs only if E or F occurs. From Statement (ii), F occurs only if B occurs and from Statement (iv), E occurs only if C occurs it means if J occurs either B or C must have occurs.

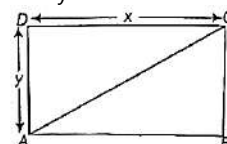
70. (c)
- $$\begin{array}{ll} R \xrightarrow{-2} P & S \xrightarrow{-2} Q \\ O \xrightarrow{+2} P & L \xrightarrow{+2} N \end{array}$$
- $$\Rightarrow$$
- $$\begin{array}{ll} A \xrightarrow{-2} Y & O \xrightarrow{-2} M \\ S \xrightarrow{+2} U & P \xrightarrow{+2} R \\ T \xrightarrow{-2} R & P \xrightarrow{-2} N \\ & Y \xrightarrow{+2} A \end{array}$$

71. (b)
- lelibroon $\rightarrow$ yellow hat
 pleka $\rightarrow$ flower garden
froti mix $\rightarrow$ garden salad
 $\therefore$ Pleka, $\rightarrow$ flower
 yellow $\rightarrow$ leli or broon
 By option
 yellow flower $\rightarrow$ lelipleka

72. (C) $E = 6 - 9 + 8 \times \frac{3}{20}$
 By given condition,
 $E = 6 + 9 \times \frac{8}{3} - 20$
 $E = 6 + 3 \times 8 - 20$
 $E = 6 + 24 - 20$
 $E = 6 + 4 = 10$

73. (b) Let in a month of January.
 (4 times) Friday $\rightarrow$ 25, 18, 11, 4 (dates)
 (4 times) Monday $\rightarrow$ 28, 21, 14, 7 (dates)
 Then, required dates of Sunday,
 Sunday $\rightarrow$ 27, 20, 13, 6
 So, Sunday of the week did the 20th of January fall that year.
74. (b) Krishna is "father-in-law" of that girl.

75. (d) Let longer side = $x = DC$
 and shorter side = $y = AD$



Now, by condition,

$$AC = y + \frac{x}{2}$$

Now, In $\triangle ACD$,

$$AC^2 = AD^2 + CD^2 \quad (\text{by Pythagoras theorem})$$

$$\Rightarrow \left(y + \frac{x}{2}\right)^2 = y^2 + x^2$$

$$\Rightarrow y^2 + \frac{x^2}{4} + xy = x^2 + y^2$$

$$\Rightarrow \frac{x^2}{4} + xy - x^2 = 0$$



$$\Rightarrow x\left\{\frac{x}{4} + y - x\right\} = 0$$

$$\Rightarrow x\left\{y - \frac{3x}{4}\right\} = 0$$

$$\Rightarrow \frac{y}{x} = \frac{3}{4}$$

$$[\because x \neq 0]$$

76. (c) SNIP (NICE) PACE
TEAR (EAST) FAST
TRAY (RARE) FIRE
POUT (OURS) CARS
 $\therefore$ CANE (AN+TS) BATS
ANTS

77. (c) From the statements, we clearly say that the reason behind the nearsightedness of the children is caused by the visual stress required by reading and other class work.

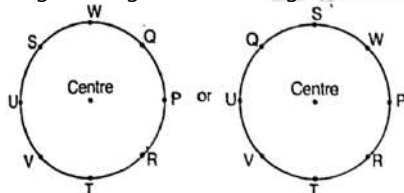
Solutions (Q.Nos. 78-80)

Randy	Vacuuming	Monday
Sally	Dusting	Tuesday
Terry	Sweeping	Wednesday
Uma	Mopping	Thursday
Vernon	Laundry	Friday

78. (d) Sweeping
79. (b) Monday
80. (c) Tuesday

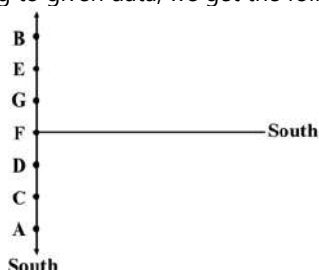
Solutions (Q.Nos. 81-82)

According to the given data, we get the following figure



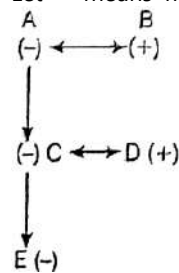
81. (a) R and V are not neighbours.
82. (d) The position of S is not fixed. So, data inadequate.
83. (c) Let the ten's place digit = x, then
By condition the unit place digit = x + 3
Now, according to question,
 $\frac{10x + (x + 3)}{4} = \frac{4}{7} \Rightarrow \frac{1 + x + 3}{2x + 3} = \frac{4}{7}$
 $\Rightarrow 1 + x + 3 = 8x + 12$
 $\Rightarrow 3x = 9$
 $\Rightarrow x = 3$

- $\therefore$ Required number = x (x + 3) = 3 (3 + 3) = 36
84. (c) After observation of given two dice, we get the number 5 is at the bottom of the dice, when number 1 is on the top.
85. (d) According to given data, we get the following figure



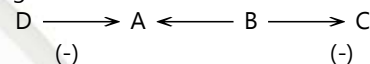
So, G is sitting third from North.

86. (a) Let '-' means 'male' and '+' means 'female'.



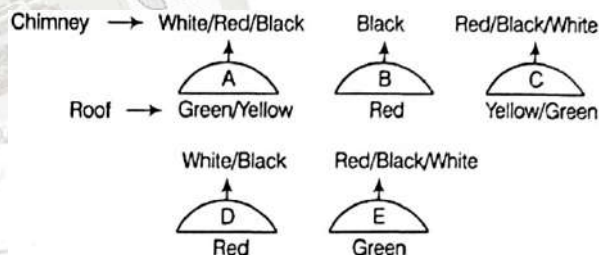
Two fathers (A, C) Two mothers (B, D)
Two sons (C, E) One father-in-law (A)
One mother-in-law (B) One daughter-in-law (D)
One grandfather (A) One grandmother (B)
One grandson (E)
So, the minimum number of persons can be 5.

87. (a) According to the directions, the relation can be solved as



So, A is brother of C.

Solutions (Q.Nos. 88-90)



88. (b) From the above diagram, it is clear that atleast two houses have red roofs.
89. (a) If house C has a yellow roof it means house D has a red roof. So, house D never has a red chimney. So, chimney of D will be of black colour, so colour of chimney of house E will be white.
90. (c) The maximum number of green roofs are 3.

ENGLISH

91.	(c)	92.	(d)	93.	(d)	94.	(c)	95.	(b)
96.	(c)	97.	(d)	98.	(d)	99.	(a)	100.	(a)
101.	(b)	102.	(d)	103.	(a)	104.	(a)	105.	(a)
106.	(d)	107.	(b)	108.	(d)	109.	(b)	110.	(a)

COMPUTER

111. (b)
An I/O processor controls the flow of information between main memory and I/O devices.
112. (d)
Magnetic tape will take highest time in taking the backup of the data from a computer.
113. (d)
ROM is a kind of secondary memory.



MAARULA CLASSES

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114. (a)
The errors that can be pointed out by compilers are syntax errors.
115. (a)
116. (b)
Required range is -128 to +127 .
117. (c)
Primary storage is fast and expensive as compared to secondary memory.
118. (a)
Control unit is used to supervise each instruction in the CPU.
119. (c)
From option (b),
Binary from $\frac{0010}{2} \frac{1111}{F} \frac{1010}{A} \frac{0000}{O} \frac{1100}{C}$
Hexadecimal
So $(2FAOC)_{16} = (00101111101000001100)_2$
From option (a).
 $(195084)_{10} = (2FAOC)_{16}$
120. (b)
 $\frac{(111)}{7} \frac{(010)}{2}_2 \rightarrow \text{Binary}$
 $\frac{7}{7} \frac{2}{2} \rightarrow \text{Octal.}$

